

PIONEER JUNIOR COLLEGE
JC2 Preliminary Examinations
In preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

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CT
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BIOLOGY

Paper 2 Structured Questions

9744/02

14 September 2018

2 hours

Candidates answer on the Question Paper.
No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your name, CT class and index number on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graph or rough working.
Do not use staples, paper clips, glue or correction fluid.

Answer **ALL** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate. You may lose marks if you do not show working or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.

Do not open this booklet until you are told to do so.

For Examiner's Use	
1	/ 9
2	/ 10
3	/ 12
4	/ 8
5	/ 9
6	/ 12
7	/ 10
8	/ 11
9	/ 8
10	/ 6
11	/ 5
Total	/ 100

Answer **all** questions in this section.

- 1 Collagen is the main structural protein in the human body. It strengthens the tendons and supports the skin and internal organs. Fig.1.1 shows the organization of collagen fibres.

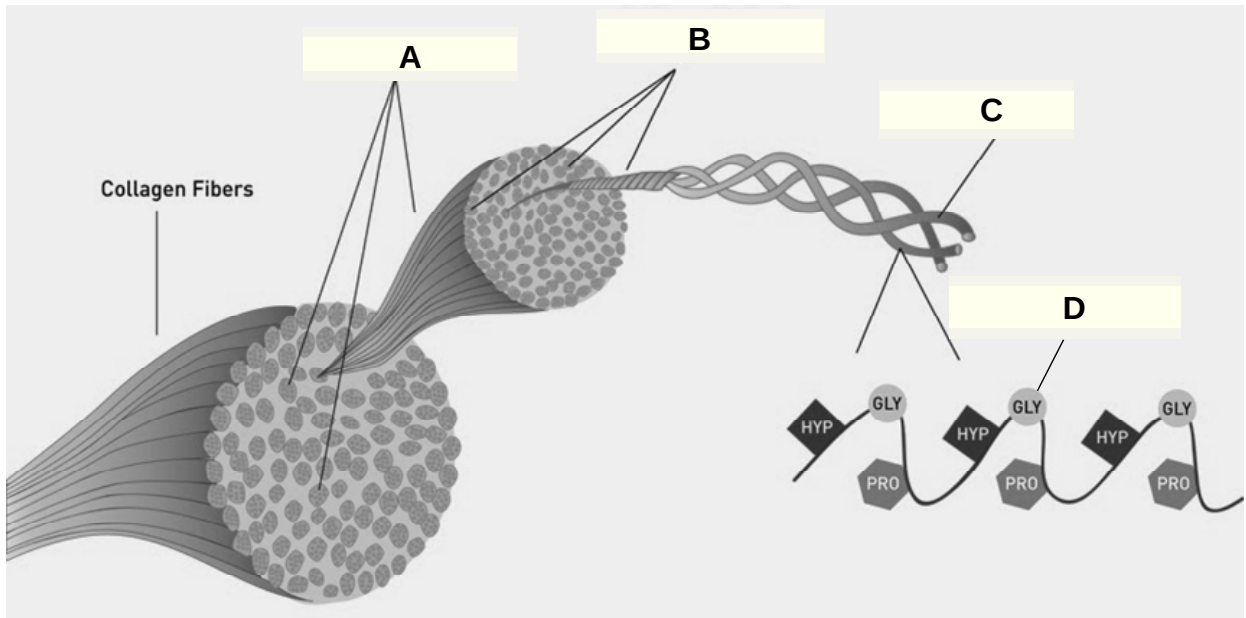


Fig. 1.1

- (a) Label structures **A**, **B**, **C** and **D** in Fig. 1.1.

A

B

C

D [2]

- (b) With reference to Fig. 1.1, describe the bonds involved in the formation of collagen which contribute to its high tensile strength.

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..... [2]

- (c) Suggest why the assembly of collagen takes place outside the cell.

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 [2]

- (d) Certain pathogenic bacteria such as *Clostridium histolyticum*, have collagenases which digest collagen tissue of their hosts, causing a form of tissue death known as gangrene.

The action of collagenase can be seen in Fig.1.2, illustrating the specificity of the active site in binding to a segment of collagen and eventually cleaving it into 2 segments.

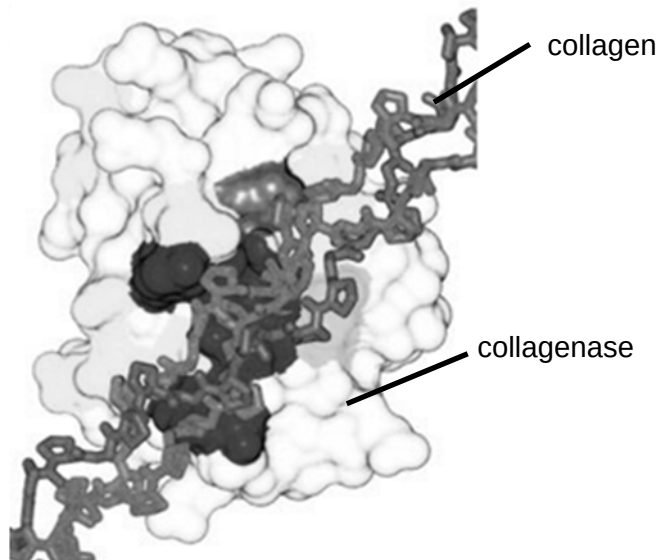


Fig. 1.2

- (i) Explain the function of amino acid residues situated at the active site of collagenase.

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 [2]

[Turn Over]

- (ii) A scar is an area of fibrous tissue that replaces normal skin after an injury. All scarring is composed of the same collagen as the tissue it has replaced, but the composition of the scar tissue, compared to the normal tissue, has slightly different arrangement. Scar tissue also lacks elasticity unlike normal tissue which distributes fiber elasticity. The extend of scarring depends on the amounts of collagen expressed at the injury site.

Santyl® Ointment is an enzymatic ointment which contains collagenase. The enzyme collagenase is derived from *Clostridium histolyticum*.

Suggest a therapeutic use of collagenase.

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..... [1]

[Total: 9]

- 2 Fig. 2.1 shows an incomplete diagram of the fluid mosaic model of membrane structure. The diagram shows the cell surface membrane of a eukaryotic cell.

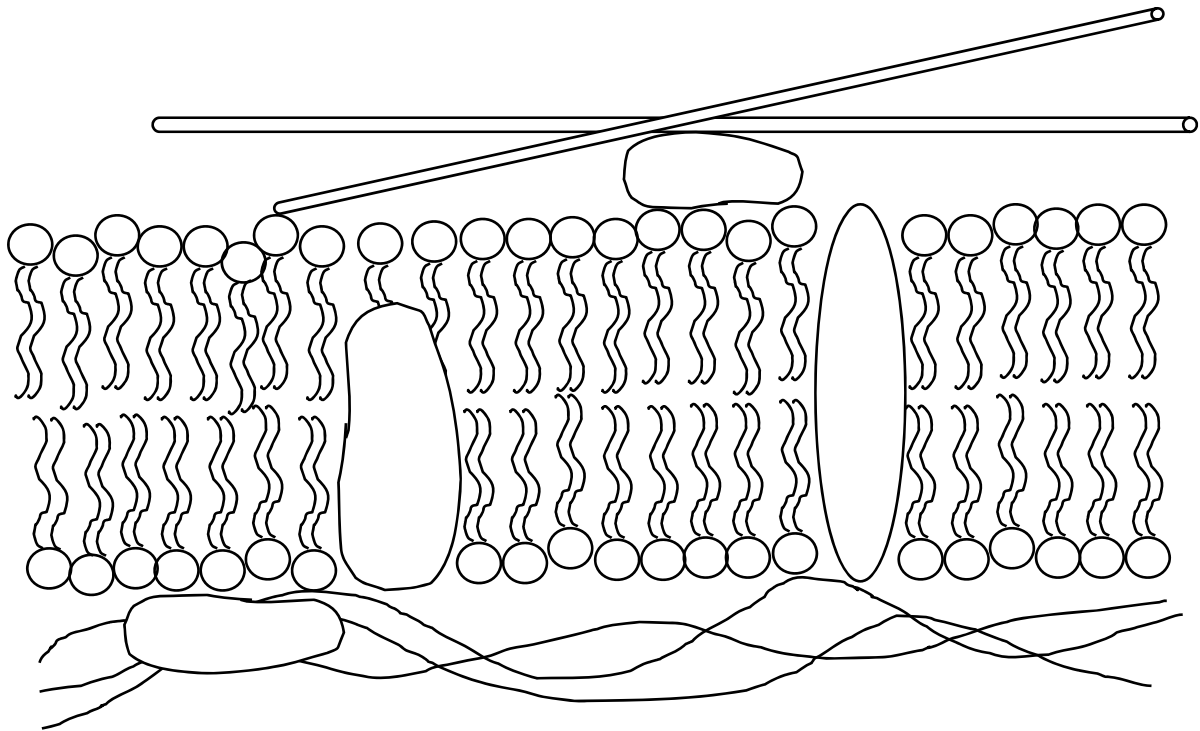


Fig. 2.1

- (a) State what is meant by fluid mosaic model.

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..... [2]

- (b) Phospholipids are a type of lipid. Lipids, in general, are made up of glycerol and fatty acids monomers covalently bonded together. Name the covalent bond and describe the breakage of this bond.

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..... [2]

[Turn Over

- (c) List four features of cell surface membranes of eukaryotic cells that are **not** visible in Fig. 2.1. and outline their roles in the cell surface membrane.

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..... [4]

- (d) The inner and outer membrane of the mitochondrion differ in the detail of their membrane components. The inner membrane is also much less permeable than the outer membrane.

Suggest **two** ways in which the structure of the inner membrane is different from that of the outer membrane to produce a **less permeable** inner membrane.

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..... [2]

[Total: 10]

- 3** The central dogma of molecular biology describes the flow of genetic information from DNA to messenger RNA (mRNA) to protein.

- (a)** A molecule involved in the flow of genetic information from mRNA to protein is transfer RNA (tRNA). Outline the role of tRNA in the production of a polypeptide.

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..... [2]

Each step in the flow of information from DNA to mRNA to protein provides the eukaryotic cell with a potential control point for regulating its functions by adjusting the amount and type of proteins synthesised.

Fig. 3.1 shows the regulation of gene expression of the albumin gene during transcription in eukaryotes. The albumin gene is associated with two control elements and a promoter.

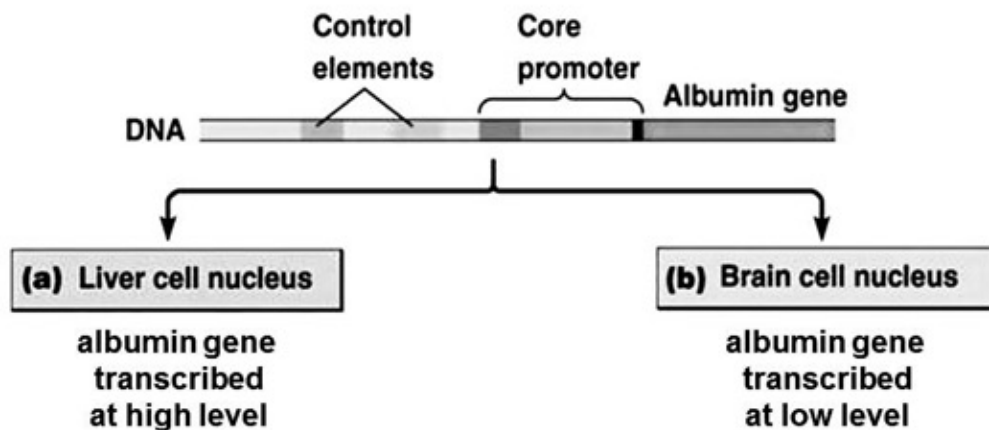


Fig. 3.1

- (b)** Explain how differential albumin gene expression in liver cells and brain cells is possible.

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..... [3]

[Turn Over]

In a normal person, the fragile X mental retardation protein (FMRP), regulates the synthesis of neuron proteins by stopping ribosomal translocation on target mRNAs. Fig. 3.2 shows how patients with fragile X syndrome (FXS) have non-functional FMRP, resulting in accelerated synaptic protein synthesis that leads to abnormal synaptic function and intellectual disability.

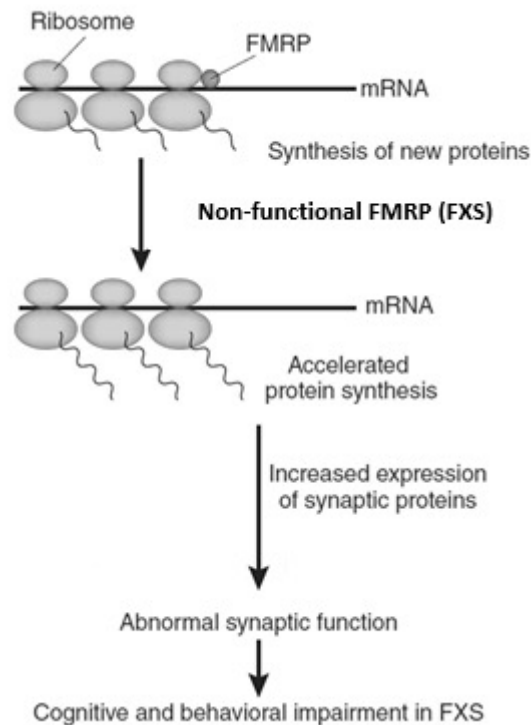


Fig. 3.2

(c) (i) State the level of control by FMRP on synaptic protein expression.

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 [1]

(ii) Describe one other control mechanism of a similar level as **(c)(i)**.

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 [2]

Research pertaining to gene regulation can contribute significantly to the treatment of genetic diseases. Spinal muscular atrophy (SMA) is a heritable motor neuron disease where patients have insufficient levels of functional survival motor neuron (SMN) protein in their motor neurons and muscle. Genetic studies have shown that all SMA patients have at least one copy of the functional SMN gene at another chromosomal locus (a result of chromosomal duplication), which is not expressed.

Scientists have shown in a study that trichostatin A (TSA), a histone deacetylase inhibitor, caused increased SMN protein levels, improved motor function and survival in mice.

Fig. 3.3 shows some results from the study. Different doses of TSA were injected in the mice and after two hours, muscle tissues were isolated. The levels of acetylated histone H3, histone H4 and SMN mRNA were measured.

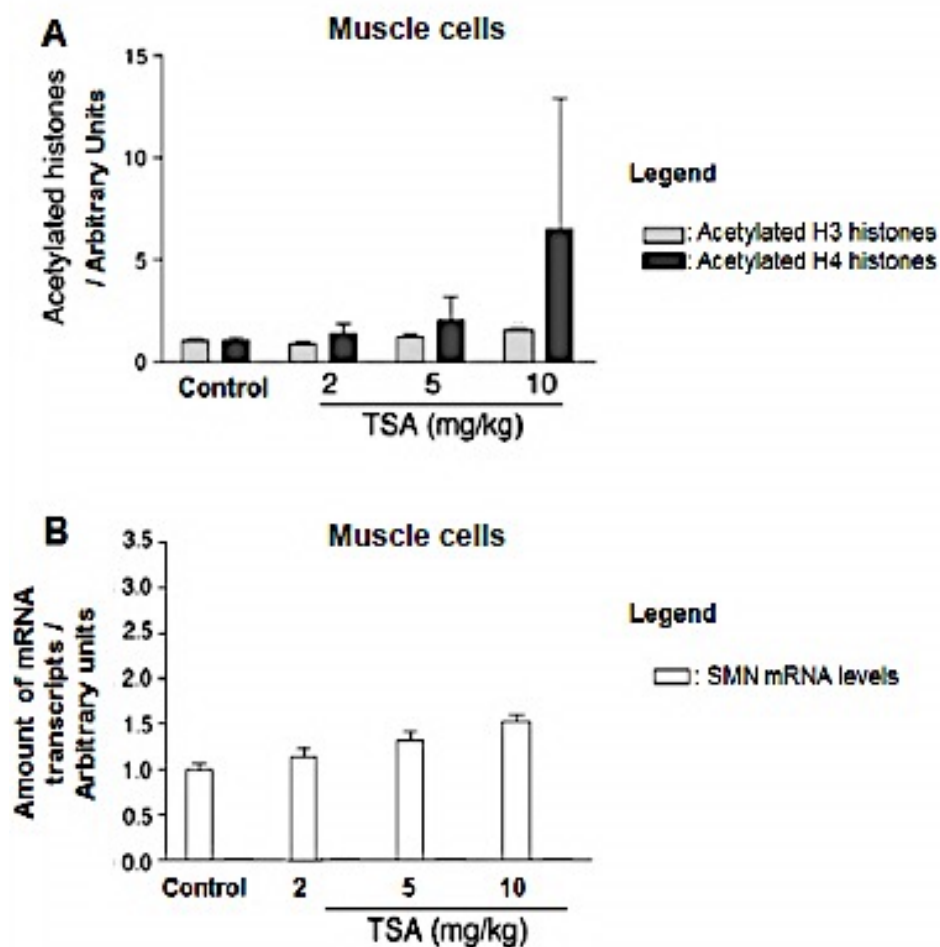


Fig. 3.3

(d) With reference to Fig. 3.3,

(i) describe the effect of TSA on the amount of SMN mRNA transcripts in muscle cells.

.....
 [1]

[Turn Over

- (ii) explain the effect of an increase in H4 histone acetylation on the regulation of transcription of the SMN gene in muscle cells.

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..... [3]

[Total: 12]

- 4 There have been many breakthroughs in stem cell research in the recent years. It has been discovered that stem cells are involved in the replacement of worn-out cells and repair of damaged tissues. Further research is being conducted to better understand the mechanism involved in controlling the behaviour of stem cells in order to better manipulate them to treat various diseases and disorders.

Stem cells undergo cell division to produce genetically identical daughter cells. Fig. 4.1 shows two cells, each at a different stage of cell division.

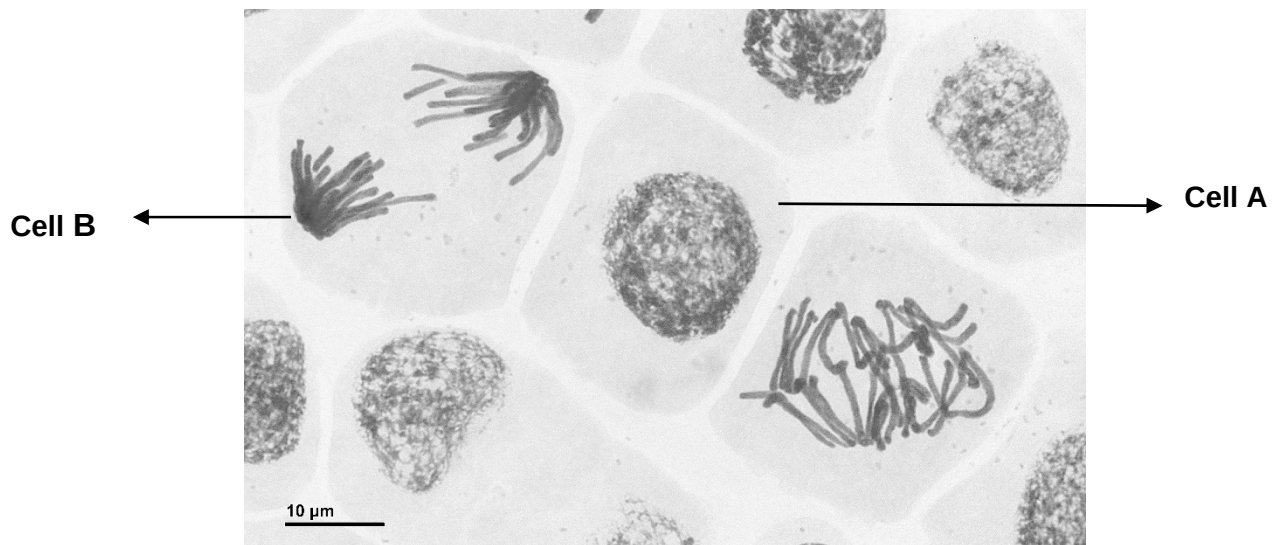


Fig. 4.1

- (a) With reference to Fig. 4.1, state the stages of cell division in Cell A and Cell B.

Cell A

Cell B [1]

[Turn over]

Fig. 4.2 shows information about the movement of chromatids in a cell that has just started metaphase of mitosis.

Key

----- = distance between chromatids

———— = distance between each chromatid and the pole to which it is moving

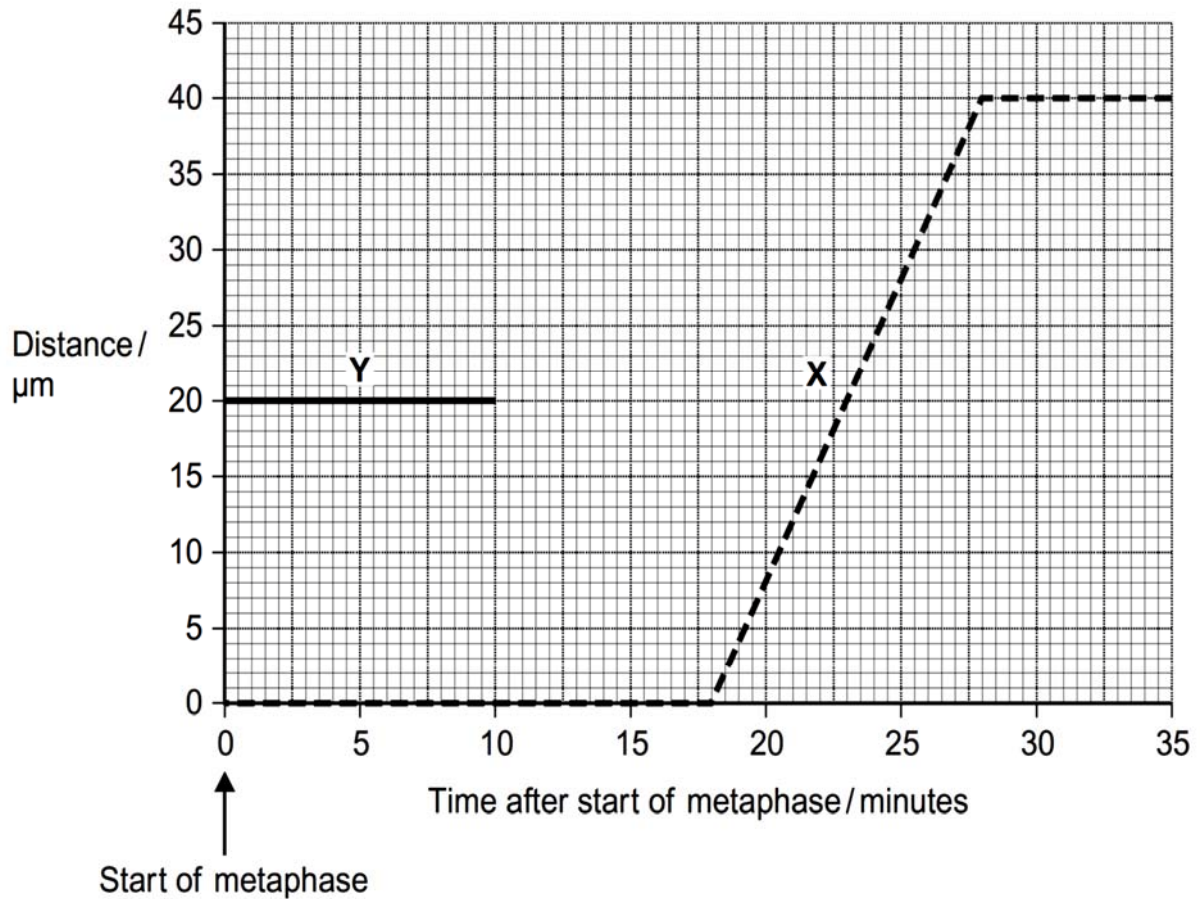


Fig. 4.2

(b) (i) With reference to Fig. 4.2, state the duration of metaphase in the cell.

..... [1]

(ii) Complete line Y on the graph.

[1]

(iii) Account for your answer in **(b)(ii)**

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..... [3]

The movement of chromatids is dependent on spindle fibres, which are made up of many tubulin subunits. Spindle fibres are lengthened at one end during mitosis by the polymerisation of tubulin subunits through GTP hydrolysis.

(c) Contrast between the structure of tubulin with that of DNA.

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..... [2]

[Total: 8]

[Turn over

5 Cancer cells do not heed the normal signals that regulate cell cycle.

(a) Describe the development of cancer as a multi-step process.

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..... [3]

Fig. 5.1. shows a cell cycle-inhibiting pathway involving the p53 protein in a normal cell.

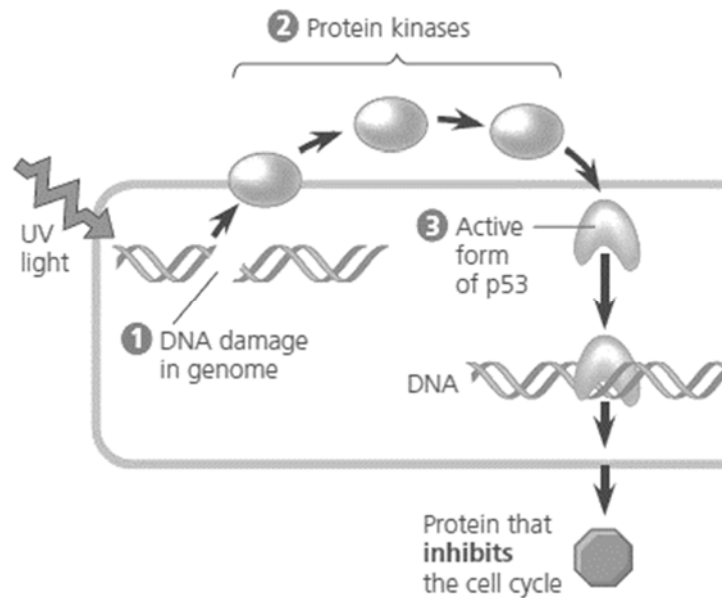


Fig. 5.1

(b) With reference to Fig. 5.1, explain how a missense mutation in p53 protein increases the likelihood of a cell becoming cancerous.

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..... [2]

- (c) Explain why mutations in the p53 gene are considered to be recessive.

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..... [2]

When a particular retrovirus that does not carry oncogenes infects a particular organism, the amount of mRNA transcribed from a particular proto-oncogene became elevated approximately 20-fold compared with uninfected individuals.

- (d) Suggest an explanation for the above observation.

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..... [2]

[Total: 9]

[Turn over

- 6 The fruit fly, *Drosophila melanogaster*, is widely used in genetic research. It has many phenotypic variants in features such as body colour, wing shape and eye colour.

Two variations from the normal-winged, grey-bodied phenotype are:

- vestigial (very short) wings, coded for by the recessive allele of the gene **N/n**
- ebony (black) body colour, coded for by the recessive allele of the gene **G/g**.

- (a) Using the symbols given, state the possible genotypes of normal-winged, grey-bodied fruit flies.

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 [2]

- (b) Describe how you would determine the genotype of a normal-winged, grey-bodied fly.

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 [3]

- (c) One of the genes for eye colour is carried on the X chromosome. This gene has different alleles coding for

- red eyes
- brown eyes
- white eyes.

The allele for red eyes (R) is dominant to the allele for brown eyes (b) and dominant to the allele for white eyes (w). The allele for brown eyes is dominant to that for white eyes.

Using these symbols, draw a genetic diagram to show how a cross between a white-eyed male fruit fly with a red-eyed female fruit fly will produce male and female offspring that are either red-eyed or brown-eyed.

Genetic diagram

[4]

[Turn over

- (d) The “eyeless” gene is a master control gene that directs the growth and development of the eyes. In flies homozygous for the mutant allele of “eyeless”, the body is not instructed to make eyes during development, resulting in a blind fruit fly as shown in Fig. 6.1. Thus when ‘eyeless’ gene is mutated, there is no expression of the gene for eye colour even though the fly may carry two copies of the gene for eye colour.

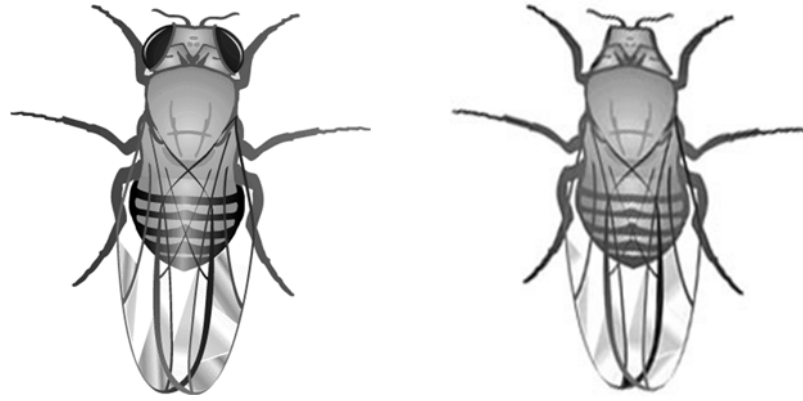


Fig. 6.1 Wild type fruit fly (left), blind fruit fly (right)

State the name for this type of interaction between “eyeless” gene and the gene for eye colour.

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 [1]

- (e) A statistical test was performed to investigate whether there was a significant difference in the mean lifespan between wild type and blind fruit flies. A summary of the results is shown in Table 6.1.

Table 6.1

Mean lifespan of fruit flies / days		Significance of difference	Total sample size
Wild type	Blind		
45	35	$p > 0.05$	50

Comment on what the results show.

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 [2]

[Total: 12]

- 7 The uptake of radioactively-labelled carbon dioxide in chloroplast was investigated.
- Three tubes, each containing different components of chloroplasts, were exposed to light.
- The results of investigation are shown in Table 7.1.

Table 7.1

Tube	Contents	Uptake of radioactively labelled carbon dioxide/ counts per minute
A	stroma and grana	3,500
B	stroma, ATP and reduced NADP	3,400
C	stroma	300

- (a) Name the substance that combines with carbon dioxide in chloroplast.

..... [1]

- (b) Explain why the results in tube **B** are similar to those in tube **A**.

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- (c) Explain why the uptake in tube **C** was less than the uptake in tube **B**.

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[Turn Over]

- (d) Cyanobacteria are prokaryotic organisms. Chloroplasts are thought to have evolved from cyanobacteria that became incorporated into larger cells. Experiments show that free-living cyanobacteria can adapt to environmental signals in the same way as chloroplasts. Fig. 7.1 shows the absorption spectra of two species of cyanobacteria

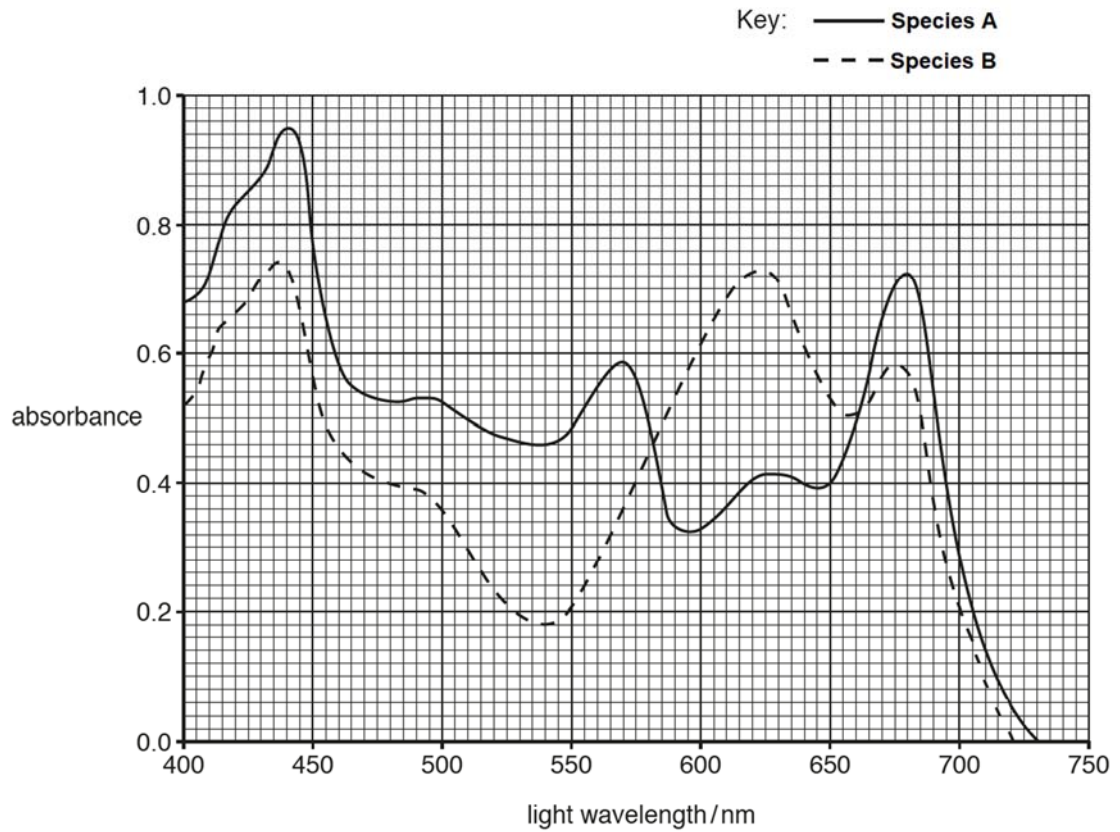


Fig. 7.1

With reference to Fig. 7.1, describe the differences in absorption spectra between the two species of cyanobacteria.

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[Total: 10]

8 A recent study of the house mouse, *Mus musculus*, on the island of Madeira resulted in the following observations.

- There are six distinct populations.
- The mice are associated with human settlements.
- The populations are located in different valleys separated by steep mountains.
- Each population has a different diploid number of chromosomes.

As a result of these observations it has been suggested that speciation is taking place.

Fig. 8.1 is a map of Madeira showing the distribution of the six populations.

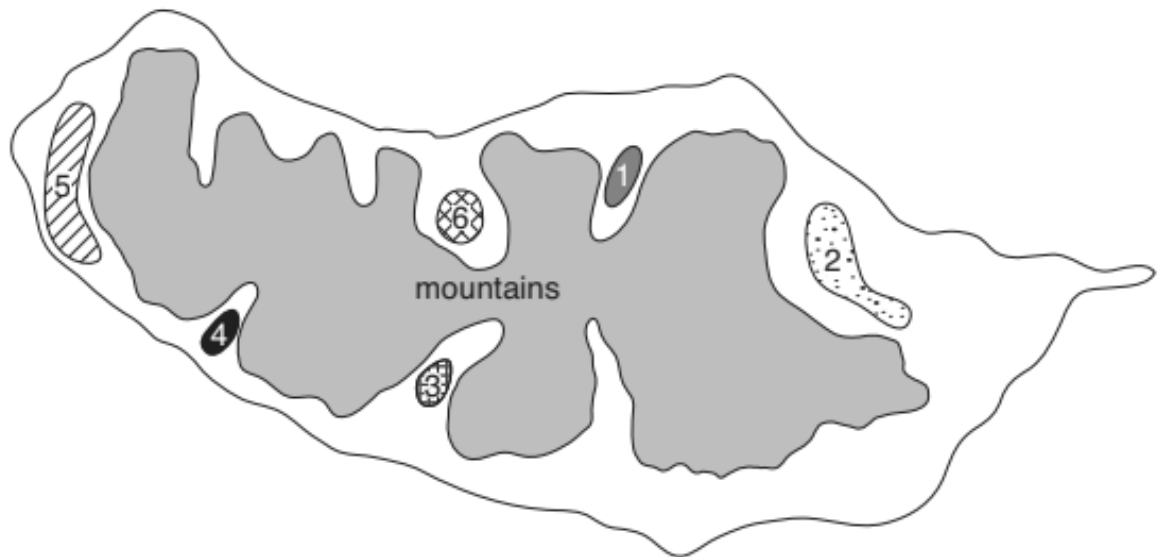


Fig. 8.1

(a) Using the information in Fig. 8.1, state the likely isolating mechanism and the type of speciation taking place.

Isolating mechanism:

Type of speciation:[2]

(b) Explain how speciation is occurring in the house mouse populations of Madeira.

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[Turn over]

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..... [5]

- (c)** Explain the likely outcome of individuals from two separate populations being mated in captivity.

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..... [2]

Based on earlier research, the native house mouse species and subspecies were distinguished based on a number of phenotypic differences such as length of the tail and fur colour. However, with modern technology, researchers have been using molecular methods such as direct DNA and amino acid comparison to further determine the phylogeny between the native house mouse species.

- (d)** Explain why molecular homology is better than anatomical homology in determining evolutionary relationship between species of native house mouse.

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..... [2]

[Total: 11]

9 Fig. 9.1 shows the glucagon signalling pathway.

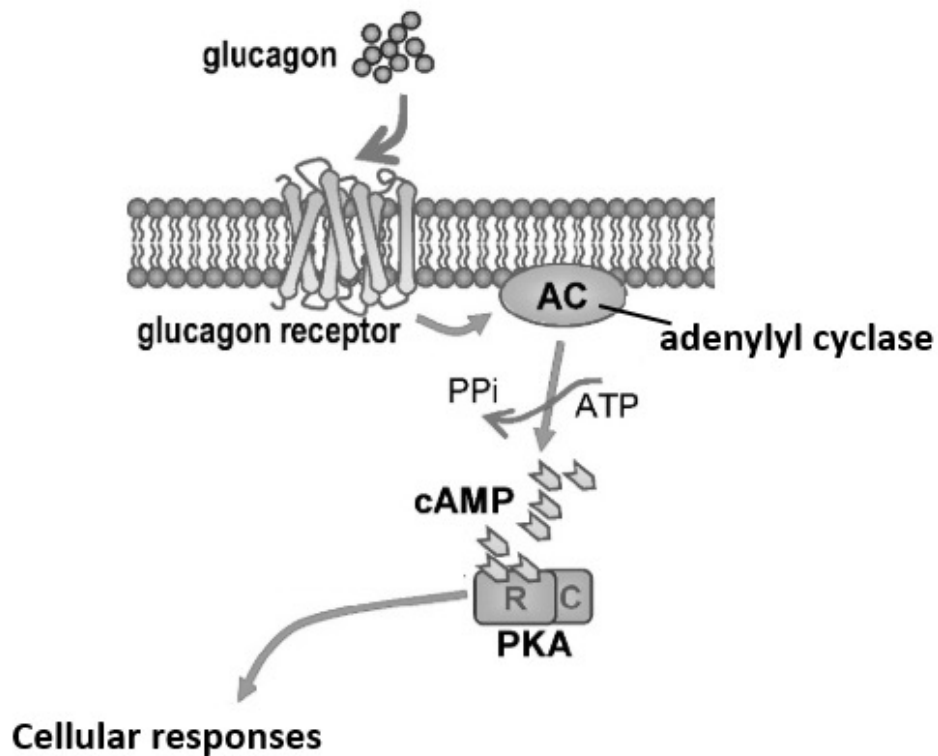


Fig. 9.1

(a) In humans, in which types of cells are glucagon receptors mainly found?

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 [1]

(b) Describe how the glucagon receptor transmits information from the external environment to activate adenylyl cyclase inside of the cell.

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 [3]

[Turn over]

- (c) Describe two cellular responses resulting from the cell signalling pathway shown in Fig. 9.1.

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..... [2]

- (d) In signal transduction pathways, how can the response of the target cell to a hormone be amplified?

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..... [2]

[Total: 8]

- 10** Our immune system needs to respond promptly to infections by pathogens. These pathogens are usually bacteria and viruses. Both our innate and adaptive immune systems are essential in helping our bodies fight against bacterial pathogens.

(a) Name one type of cell in the innate immune system and state its role in eradicating bacterial infections.

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 [1]

Cholera is an infectious disease caused by the bacterium, *Vibrio cholerae*. Most people who have recovered from cholera rarely become ill again from the disease. In these people, antibodies have been identified that will bind either to the cholera toxin, or to the bacterium flagellum, or to the main bacterial cell.

(b) Explain why the antibodies are different, each one specific to its target.

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 [3]

Fig. 10.1 shows part of the process of the production of a heavy chain polypeptide for an antibody. At the top of the diagram is the chromosomal arrangement found in an immature B cell.

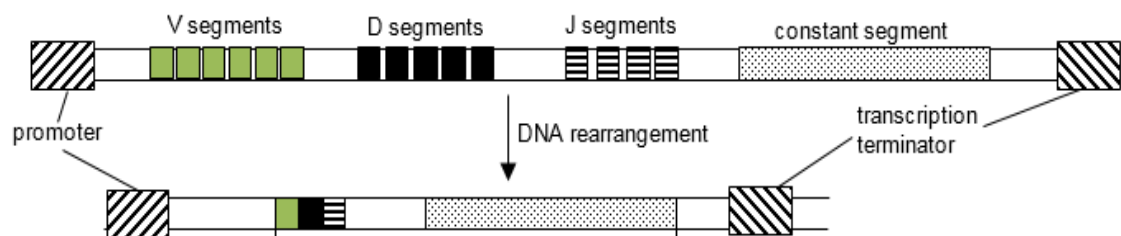


Fig 10.1

[Turn over

- (c) Identify the process represented in Fig. 10.1 and state the enzyme involved in the process.

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..... [1]

Vaccinations for some diseases are available. The development of the influenza vaccine is a dynamic research process as frequent genetic changes in the surface antigens of the virus renders each vaccine less effective over time. However, another mechanism of genetic change that results in flu pandemics can also occur irregularly and unpredictably. This occurs when two different strains of the influenza virus infect the same host cell simultaneously.

- (d) Suggest why the influenza virus can undergo this irregular and unpredictable mechanism of genetic change.

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..... [1]

[Total: 6]

- 11** Surface ocean carbon dioxide concentration can be determined by recording the concentration of carbon dioxide, in a closed volume of air that was circulated with a constantly renewed supply of water obtained two to three meters below the surface of the ocean.

Fig. 11.1 are graphs showing the changes in concentration of carbon dioxide in the air and changes in pH in the oceans of Bermuda and Hawaii from 1990 to 2010.

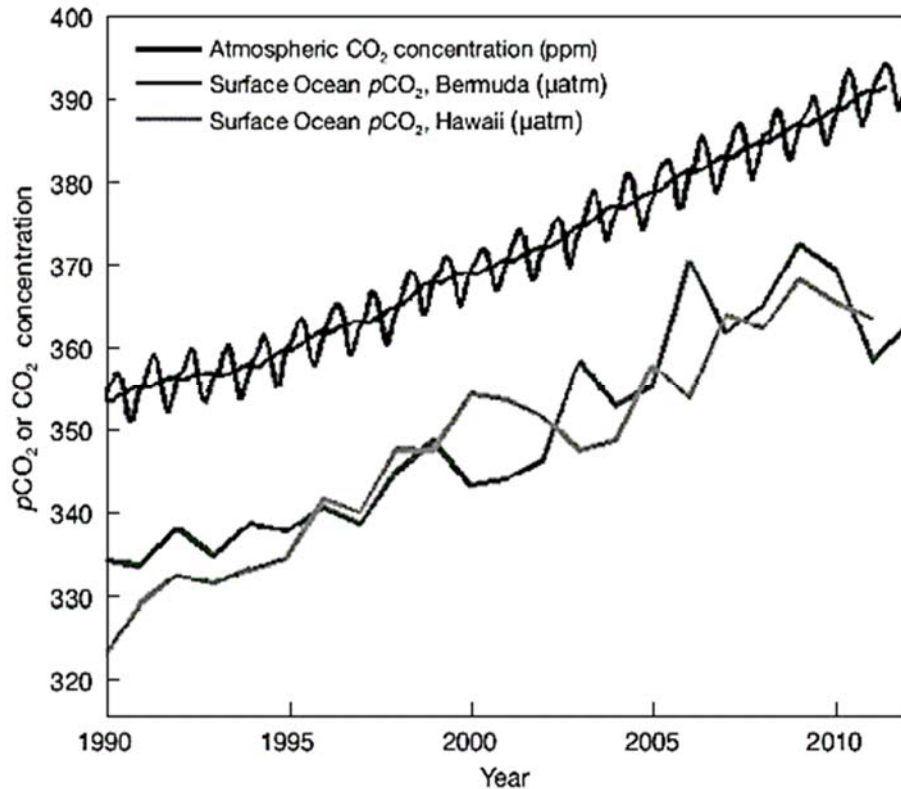


Fig. 11.1

- (a)** Briefly explain two human activities that have resulted in increased emission of carbon dioxide into the atmosphere.

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..... [2]

Increasing ocean surface carbon dioxide concentration also has negative impacts on coral reefs.

Coral reefs are at risk of damage by human activities. A study was conducted to see the effects of climate change on coral reefs.

[Turn over

Coral reef sites were subjected to two different environmental conditions i.e. exposed site and sheltered site. Coral reefs in exposed site was exposed to climate change environmental conditions. Coral reefs in the sheltered site was exposed to normal environmental conditions.

Table 11.1 shows coral cover area at exposed and sheltered sites.

Table 11.1

Experimental site		Area of healthy coral reef/m ²	Average area of healthy coral reef/m ²
Exposed Site	Site 1	120	
	Site 2	100	
	Site 3	150	
Sheltered Site	Site 1	82	
	Site 2	75	
	Site 3	69	

With reference to Table 11.1,

- (b)** Complete Table 11.1 by calculating the average area of healthy coral reef in exposed and sheltered site. [1]

Show your working

- (c)** Explain two ways how climate change damages coral reefs.

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..... [2]

[Total: 5]

END OF PAPER